

JKBOSE 2026

CLASS 11TH

PLANT KINGDOM

LECTURE-04

BOTANY

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Topics To Covered

- 1 Introduction of Bryophytes. ✓
- 2 General characters of bryophytes ✓
- 3 Classification of bryophytes ✓
- 4 Economic importance of bryophytes ✓

INTRODUCTION

- Study of Bryophytes is called "Bryology" ✓
- term Bryophyta – Coined by "Robert Brown". ✓
- Father of Indian Bryology = Prof. Shiv Ram Kashyap... ✓

GENERAL CHARACTERS OF BRYOPHYTES

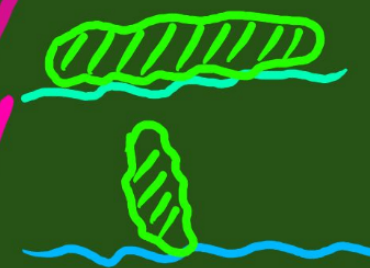
Habitat: Commonly growing in moist shaded areas in the hills.

↓
"Sciophytes" — plants growing in shaded areas.

Main plant body is: Gametophyte (n)

↓
produces Gametes (n) by mitosis.

- **Thalloid** (but more differentiated as compared to algal thallus).
- possess → 1) Root like Str. kld as **Rhizoid**
 - unicellular
 - multicellular
- 2) Stem like Str. kld as **Cauloid**
 - prostrate ✓
 - Erect ✓
- 3) leaf like Str. kld as **Phylloid**.



Note:

①:

True root
True stem
True leaves

}

-nt

in

Bryophytes

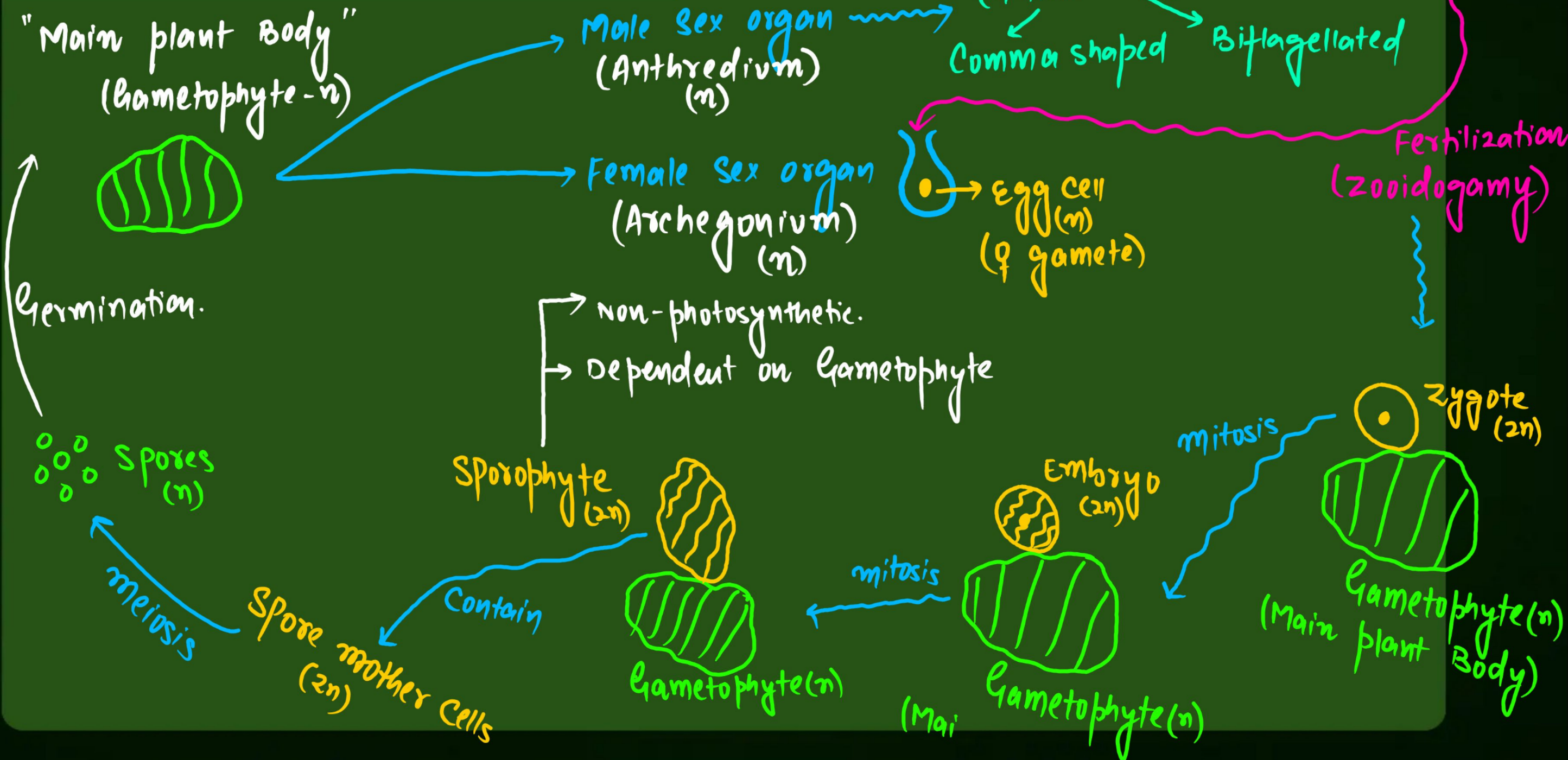
Because of

-nse of

Vascular bundle (xylem & phloem).

②. Bryophytes — avascular. ✓

General life cycle of Bryophytes:



①. Bryophytes — Sex organs $\left\{ \begin{array}{l} \text{Antheridium } (\sigma) \\ \text{Archegonium } (\text{♀}) \end{array} \right\}$ Multicellular ✓
and Jacketed. ✓
flask shaped
↓
egg cell

②. Bryophytes lives on land but requires water for fertilization i.e, why Bryophytes are AKA "Amphibians of Plant Kingdom".

③. Bryophytes — primitive Embryophytes { because Embryo is formed first time in Bryophytes in plant kingdom".

④. Sporophyte ($2n$) — is dependent on Gametophyte (n)

⑤. Sporophyte (2n) — produces spores (n) by meiosis.

CLASSIFICATION OF BRYOPHYTES

03 Classes

① liver worts
(Hepaticopsida)
↓
Syllabus (✓)

② Hornworts
(Anthocerotopsida)
↓
Syllabus - (x)

③ Mosses
(Bryopsida)
↓
Syllabus - (✓)

LIVERWORTS

Ex. Marchantia, Riccia

• Main plant Body: Gametophyte (n) — produces gametes (n) by mitosis.

possess → Root like Str. (Rhizoid)

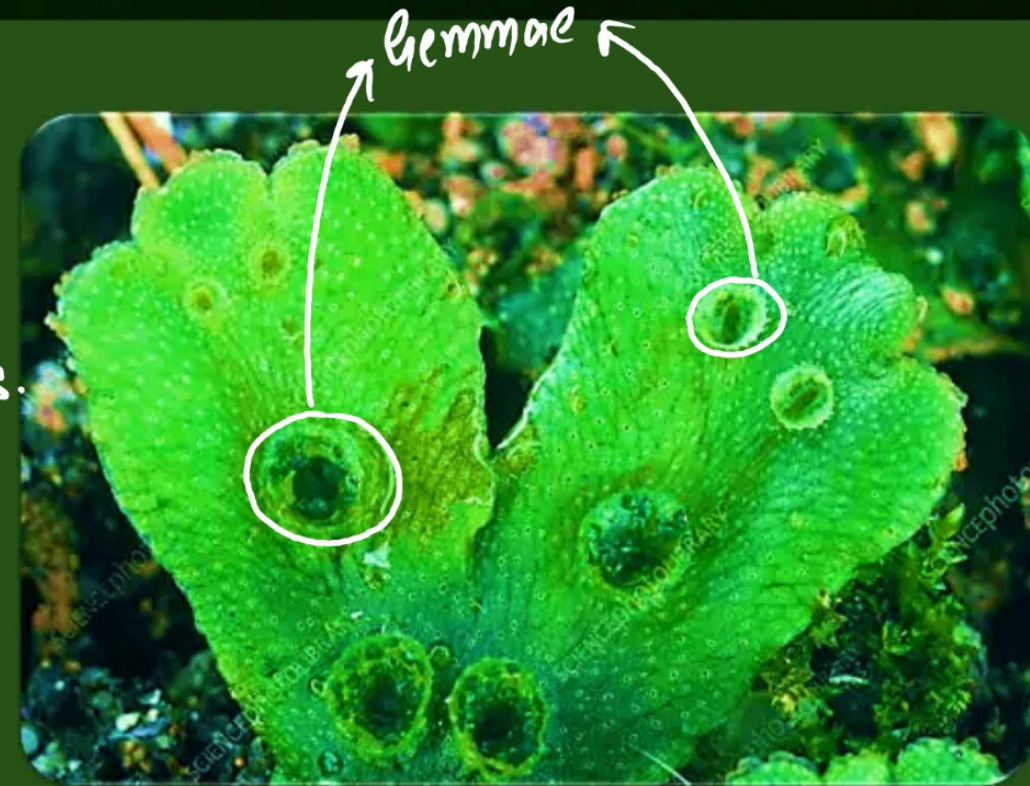
↓
"Unicellular"

→ Stem like Str. (Cauloid)

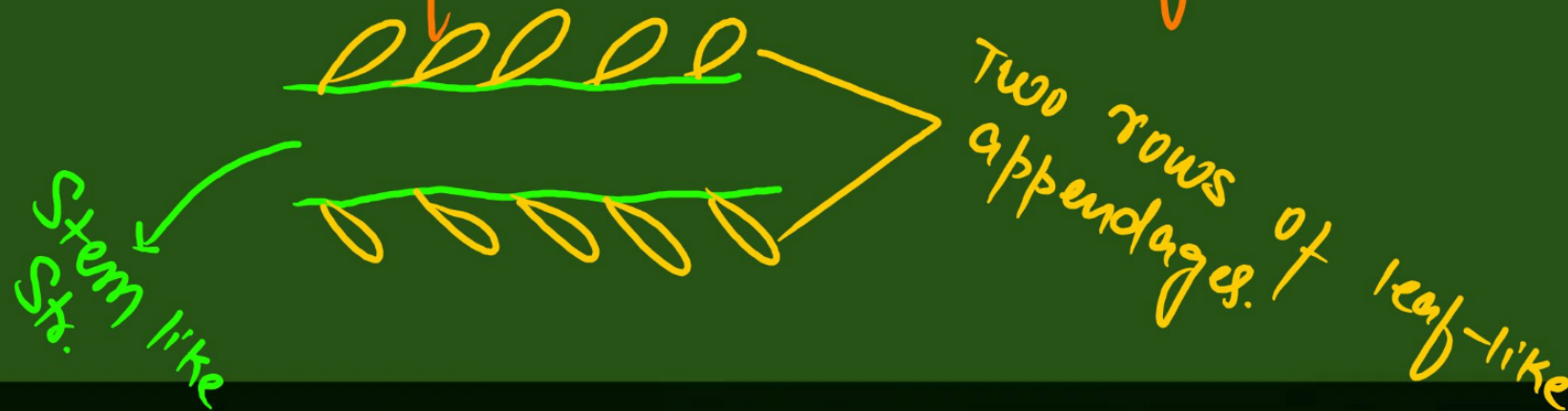
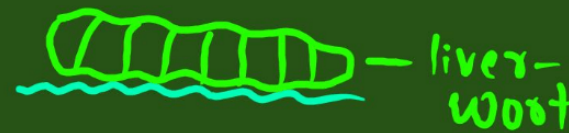
↓
"Prostate"

→ leaf like Str. (phylloid.)

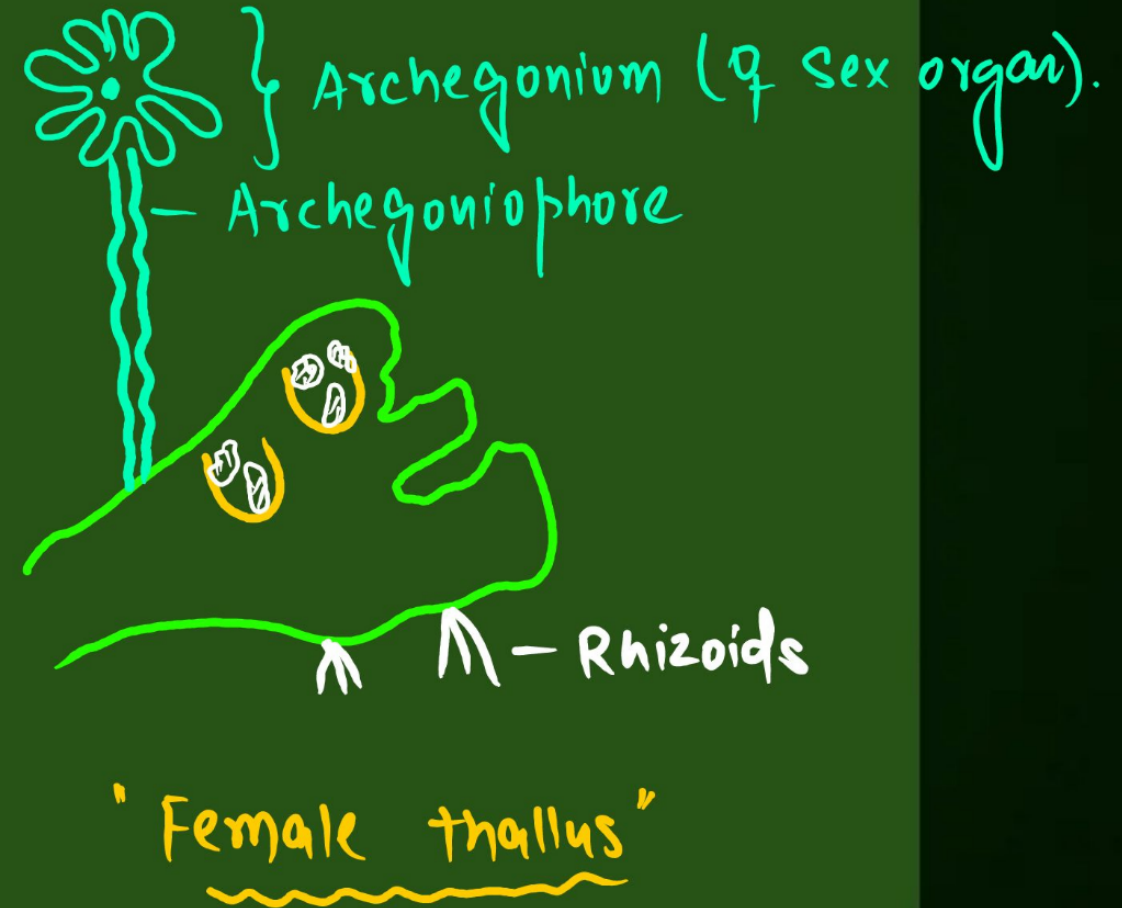
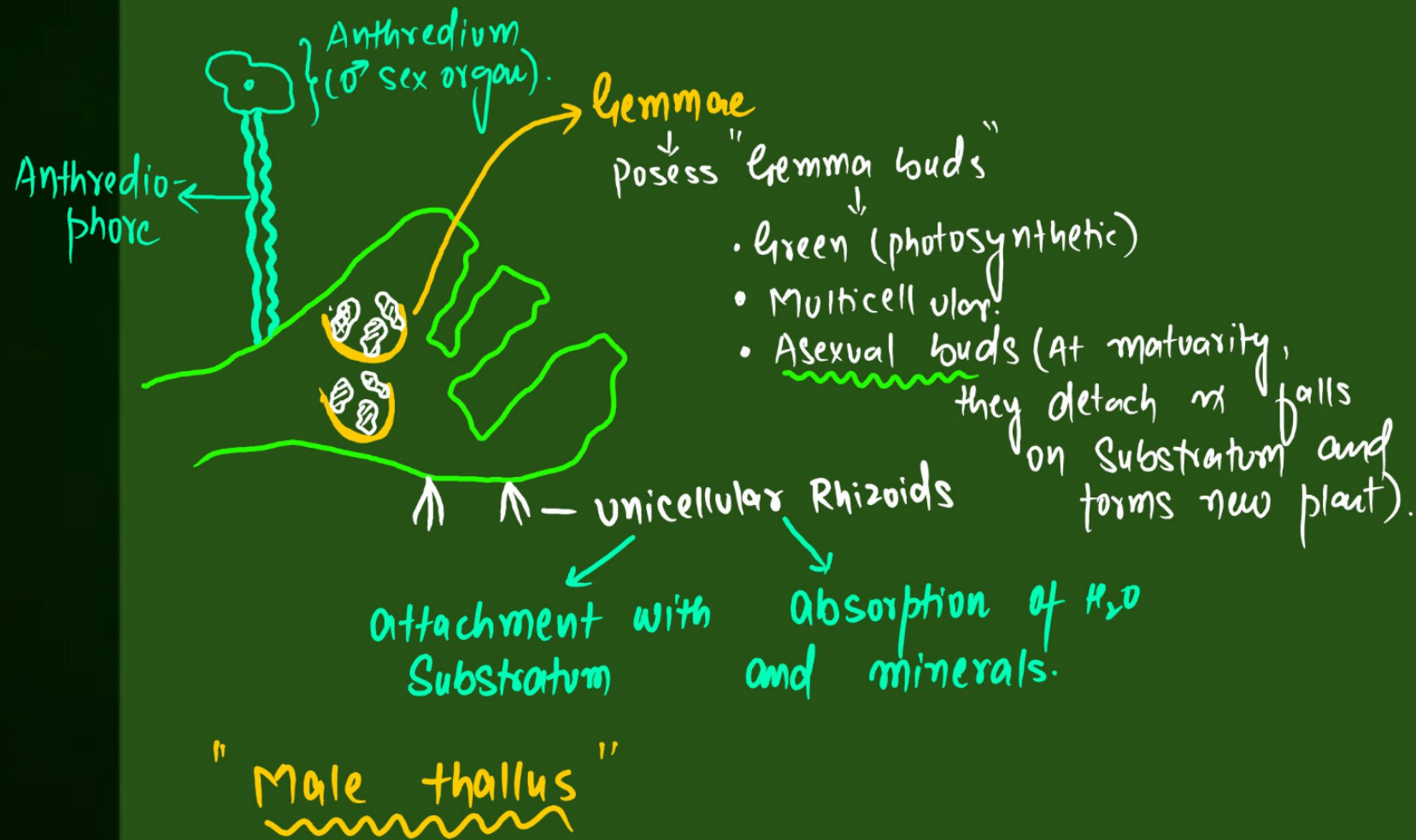
≡ In some members of liverworts — two leaf-like rows are int on Stem-like Str.



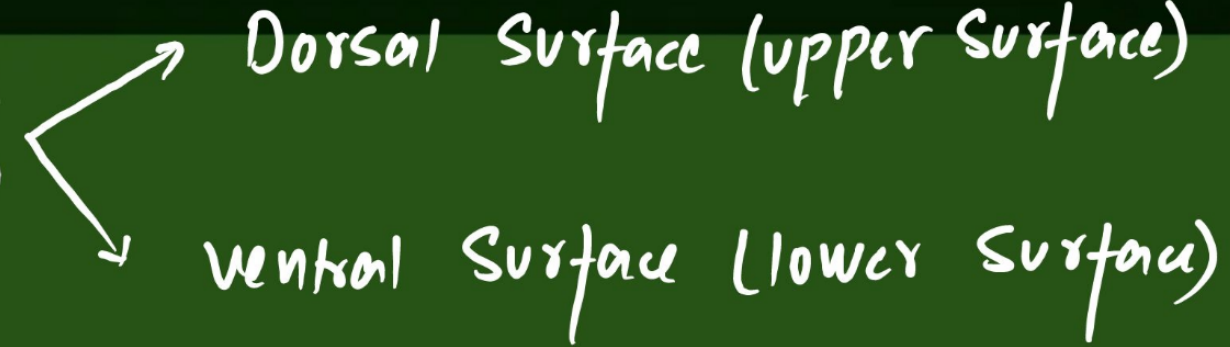
"Marchantia"



- liver worts
 - Dioecious — when ♂ and ♀ sex organs are +nt on two different individuals. ex. Marchantia*
 - Monoecious — when both ♂ and ♀ sex organs are +nt on same individuals. ex. Riccia.
- Marchantia (Dioecious)*



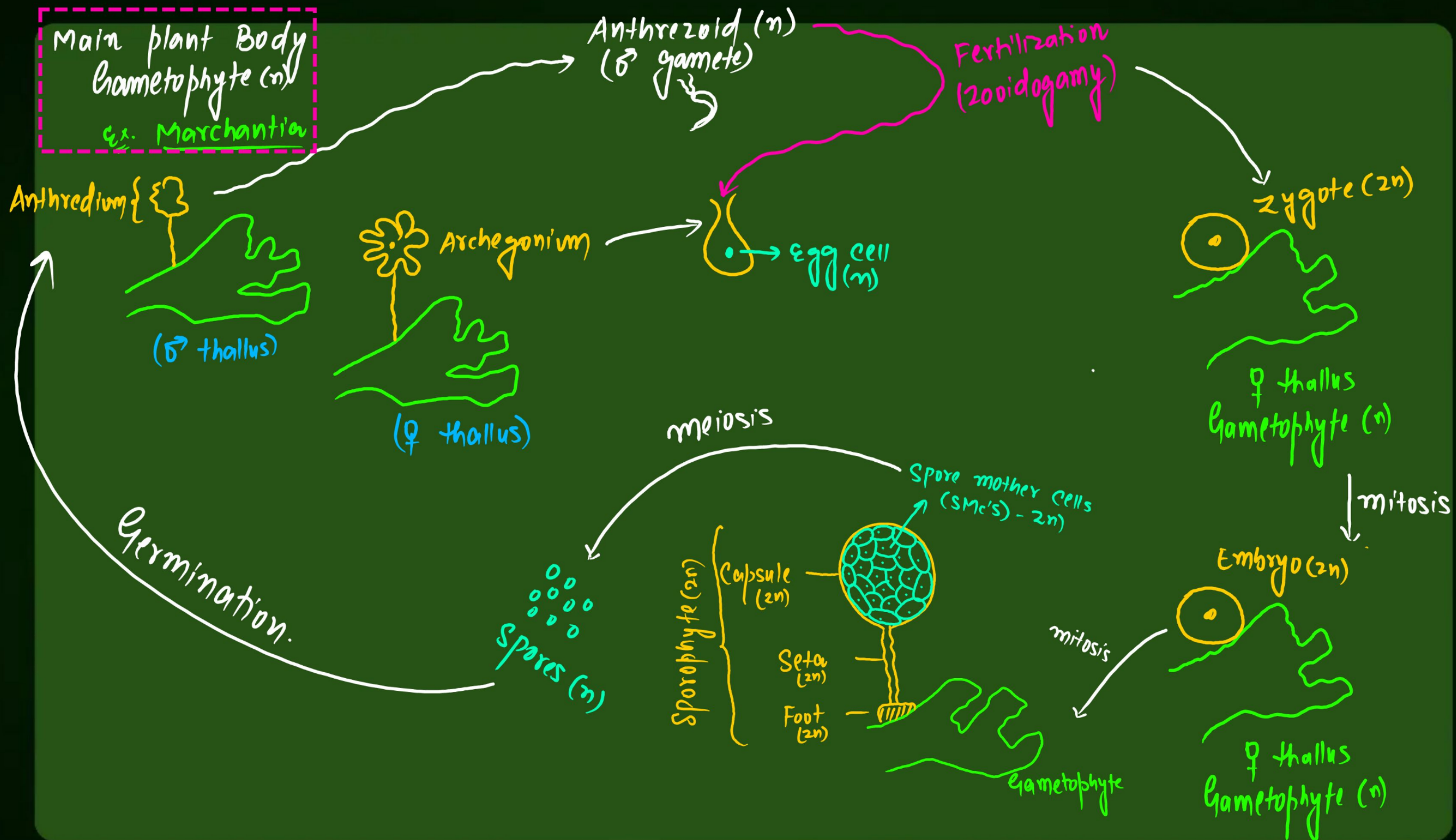
• liverworts = Dorsiventrally flattened



```
graph LR; A[Dorsiventrally flattened] --> B[Dorsal Surface (upper surface)]; A --> C[Ventral Surface (lower surface)];
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Dorsal Surface (upper surface)

Ventral Surface (lower surface)



QUESTIONS

In Bryophytes what is absent:

- ☐ **A** Embryo formation ✗
- ☐ **B** Fertilization ✗
- ☐ **C** Motile gametes ✗
- ☒ **D** True roots and vascular tissue

QUESTIONS

The similarity between bryophytes and algae is the presence of:

- ☐ **A** Thallus plant body, autotrophic nutrition and presence of vascular tissue ✗
- ☒ **B** Thallus plant body, autotrophic nutrition and lack of vascular bundle
- ☐ **C** Presence of rhizoid, cauloid and phylloid
- ☐ **D** All of the above

QUESTIONS

In bryophyta, organs are referred to as 'Leaf like' and 'Stem like' and not the true leaf and stem because

- ☒ **A** They lack vasuclar tissues ✓
- ☐ **B** They are non-green
- ☐ **C** They do not function as leaf and stem
- ☐ **D** All the above

Thank
You